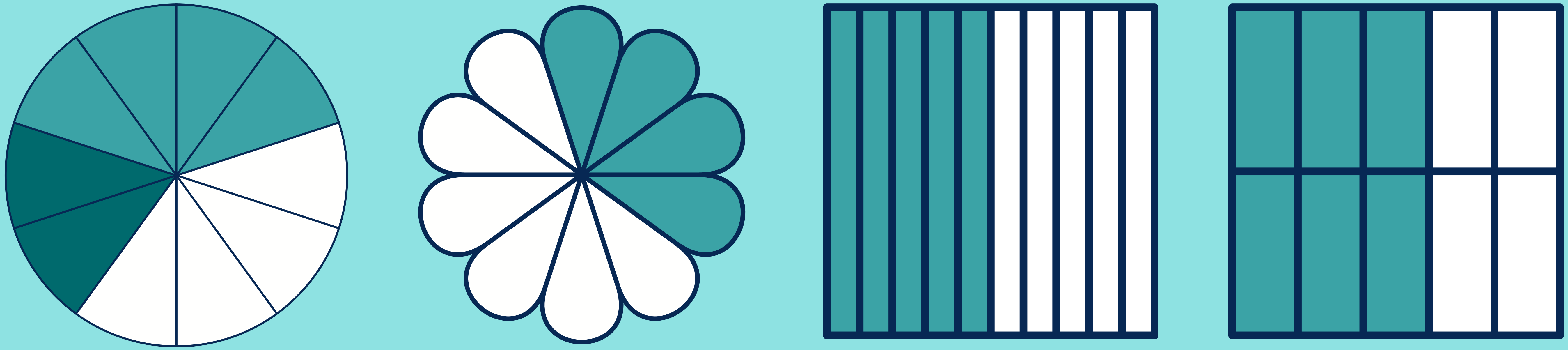
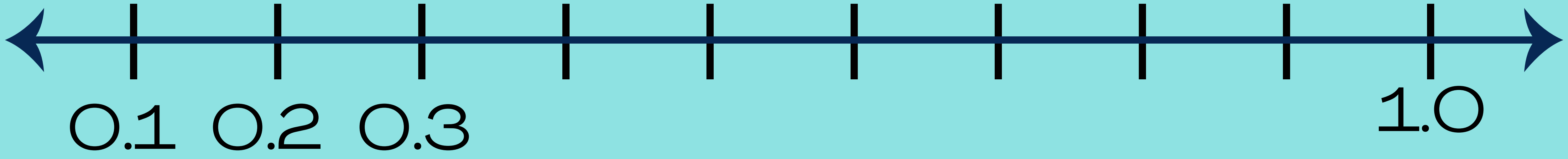
DECIMALS

Grade 3

Q: How many tenths make one whole?

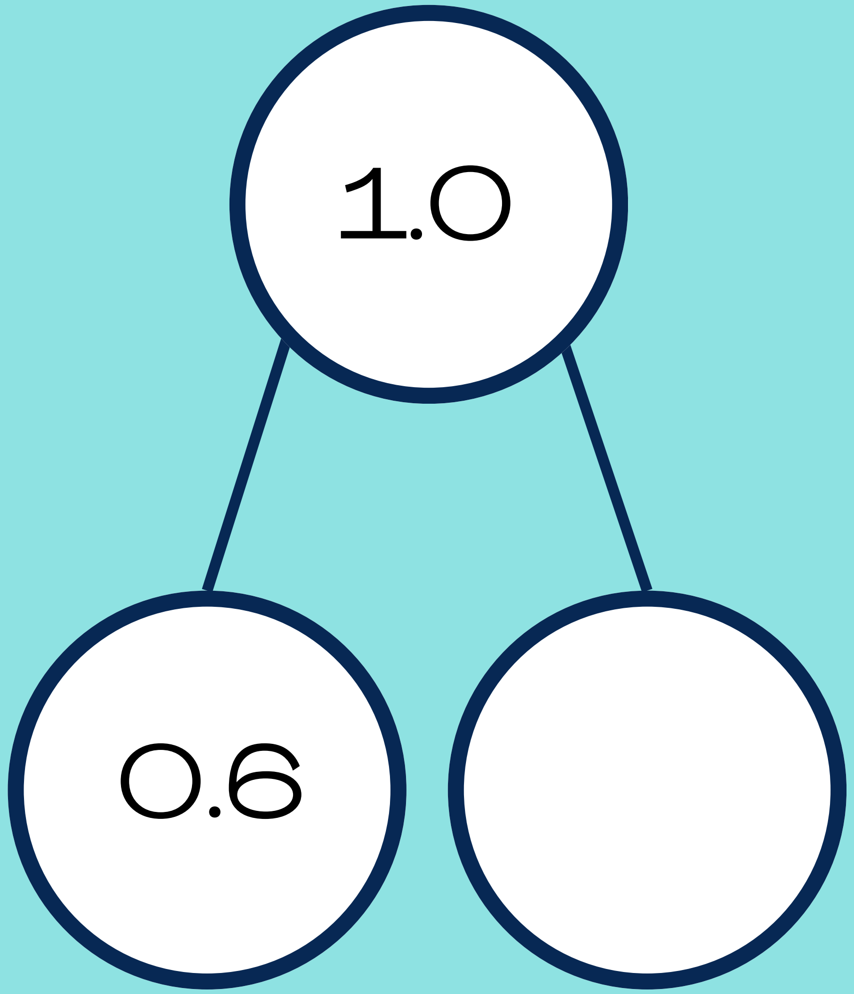




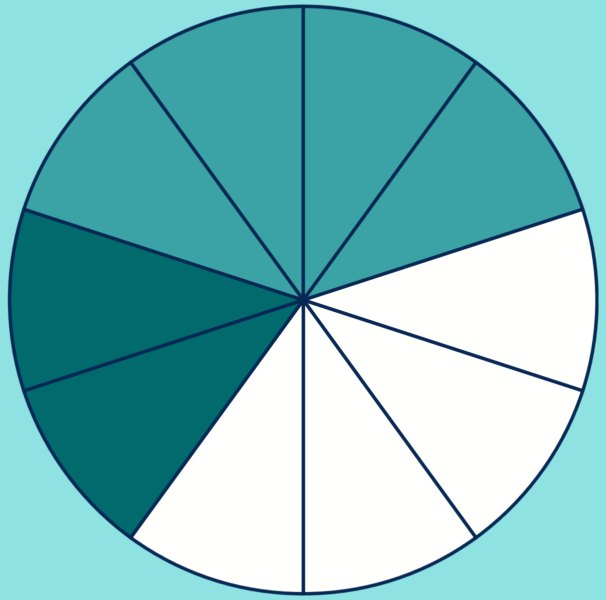
Q: How many MORE tenths are needed to make one whole?

$$0.4 + \boxed{} = 1$$

Q: How many MORE tenths make one whole?



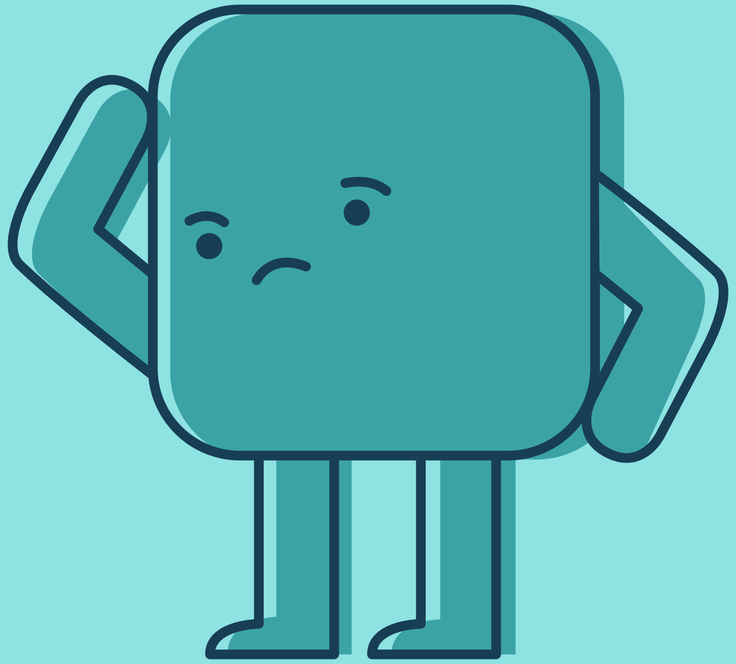
Q: How many MORE tenths make one whole?



1 whole		
0.2		

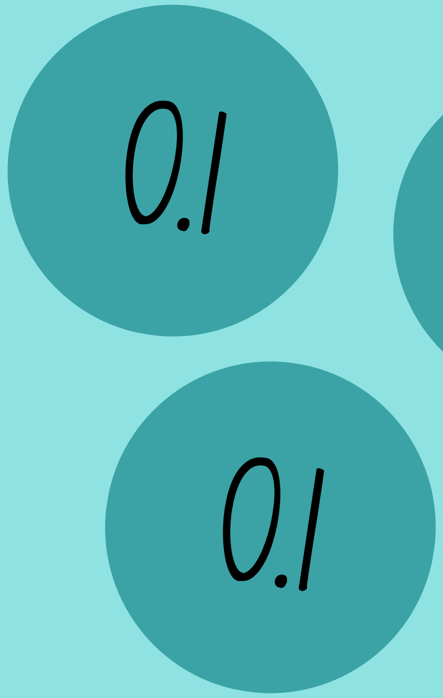
PROBLEM SOLVING

Q: Sam has four tenths of a cake and Liv has two tenths. How much cake is left? Write your answer as a fraction and a decimal.

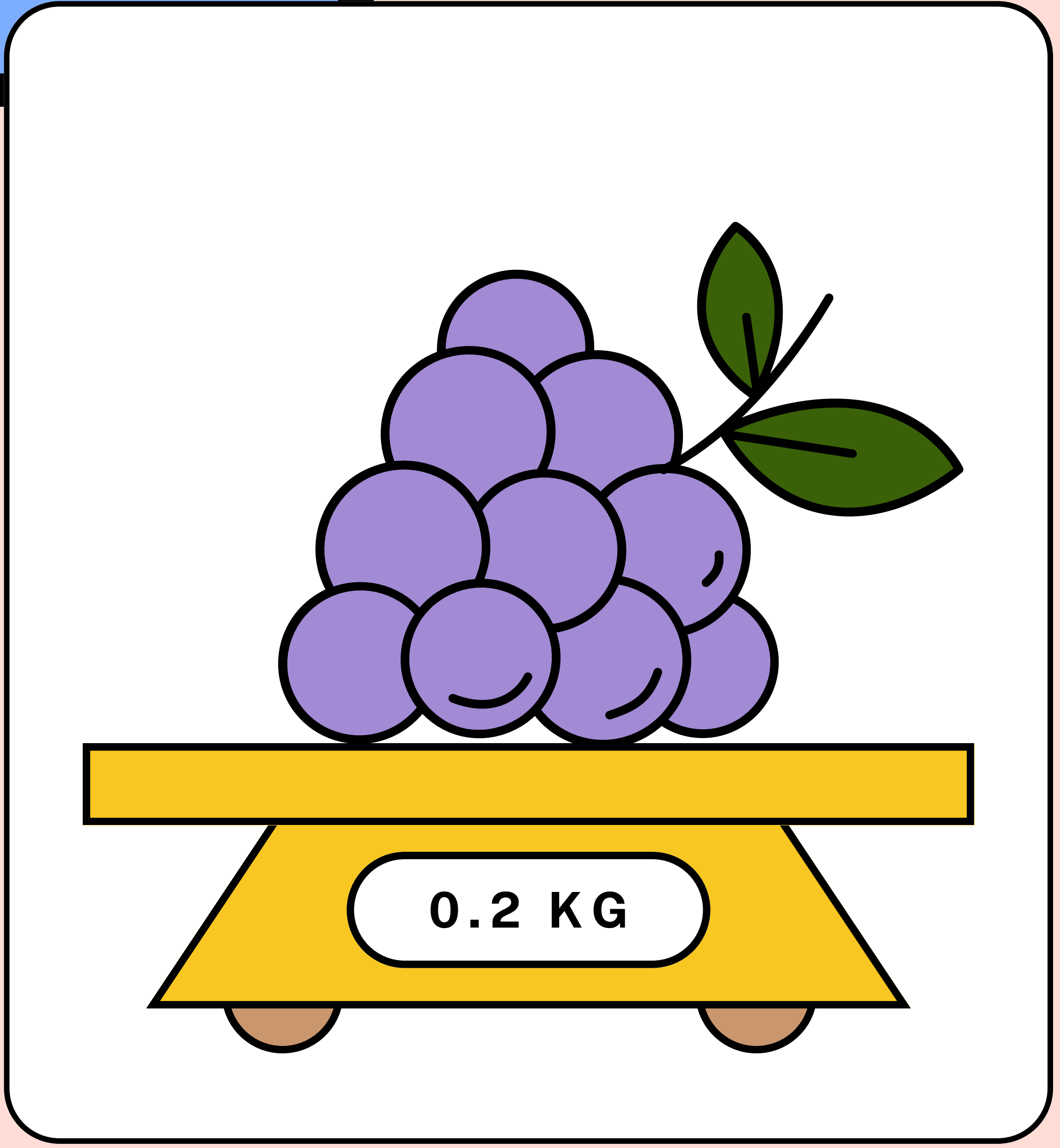


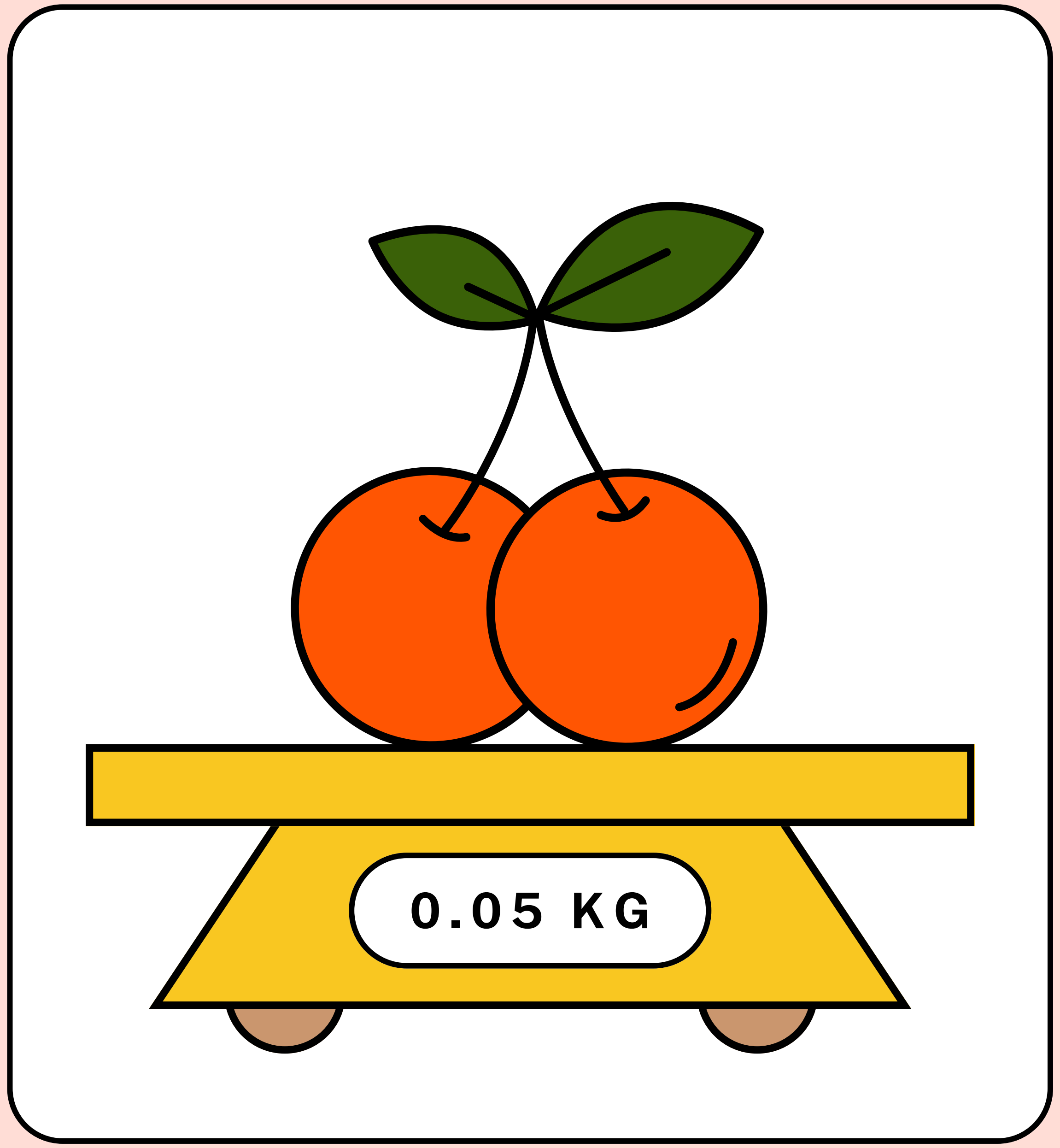
PROBLEM SOLVING

Q: Ali has 3 counters. Each counter is worth one tenth.
Grace has one more counter than Ali.
Later Ben comes to join them and altogether their counters total one whole. How many counters does Ben have?

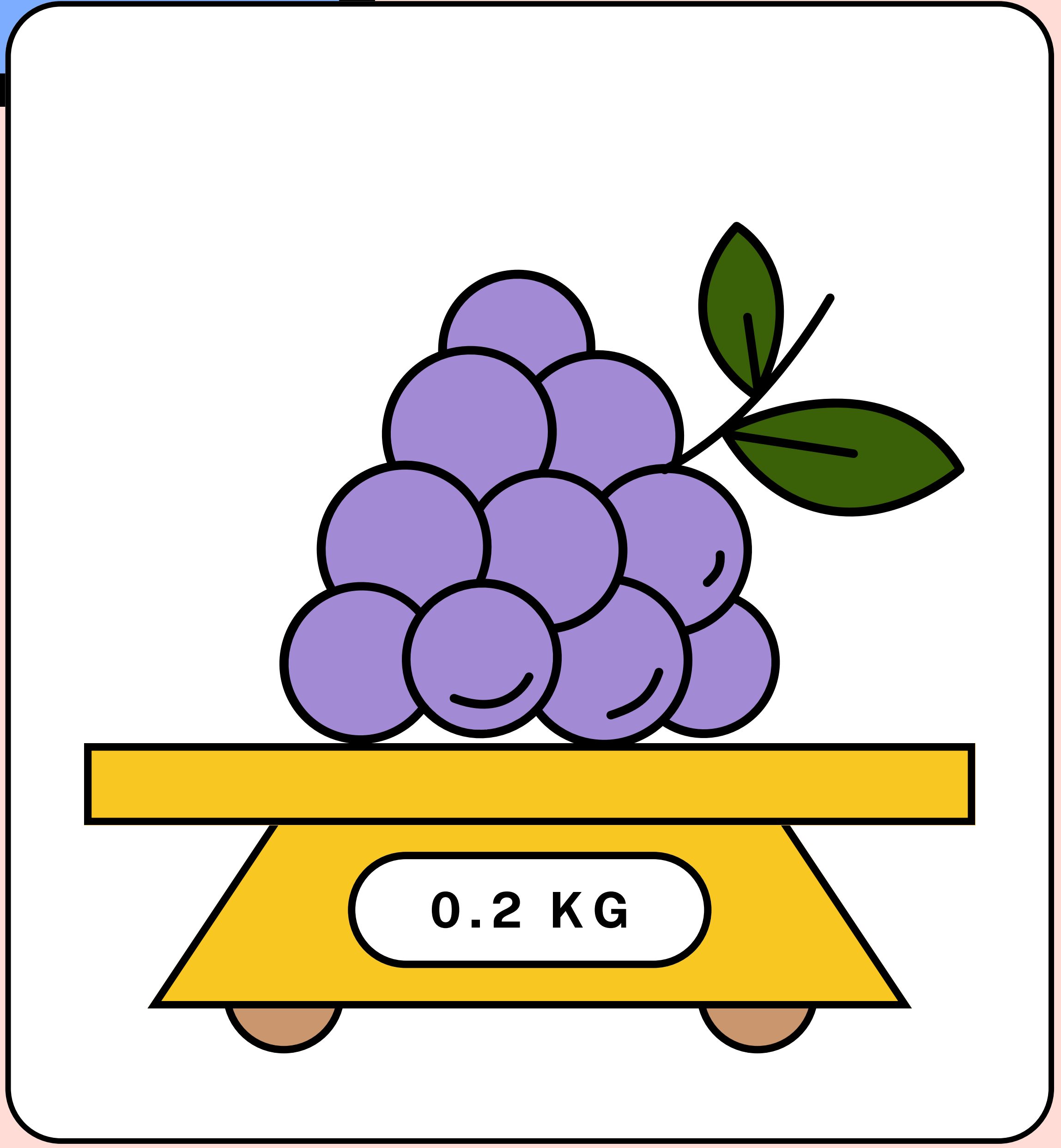


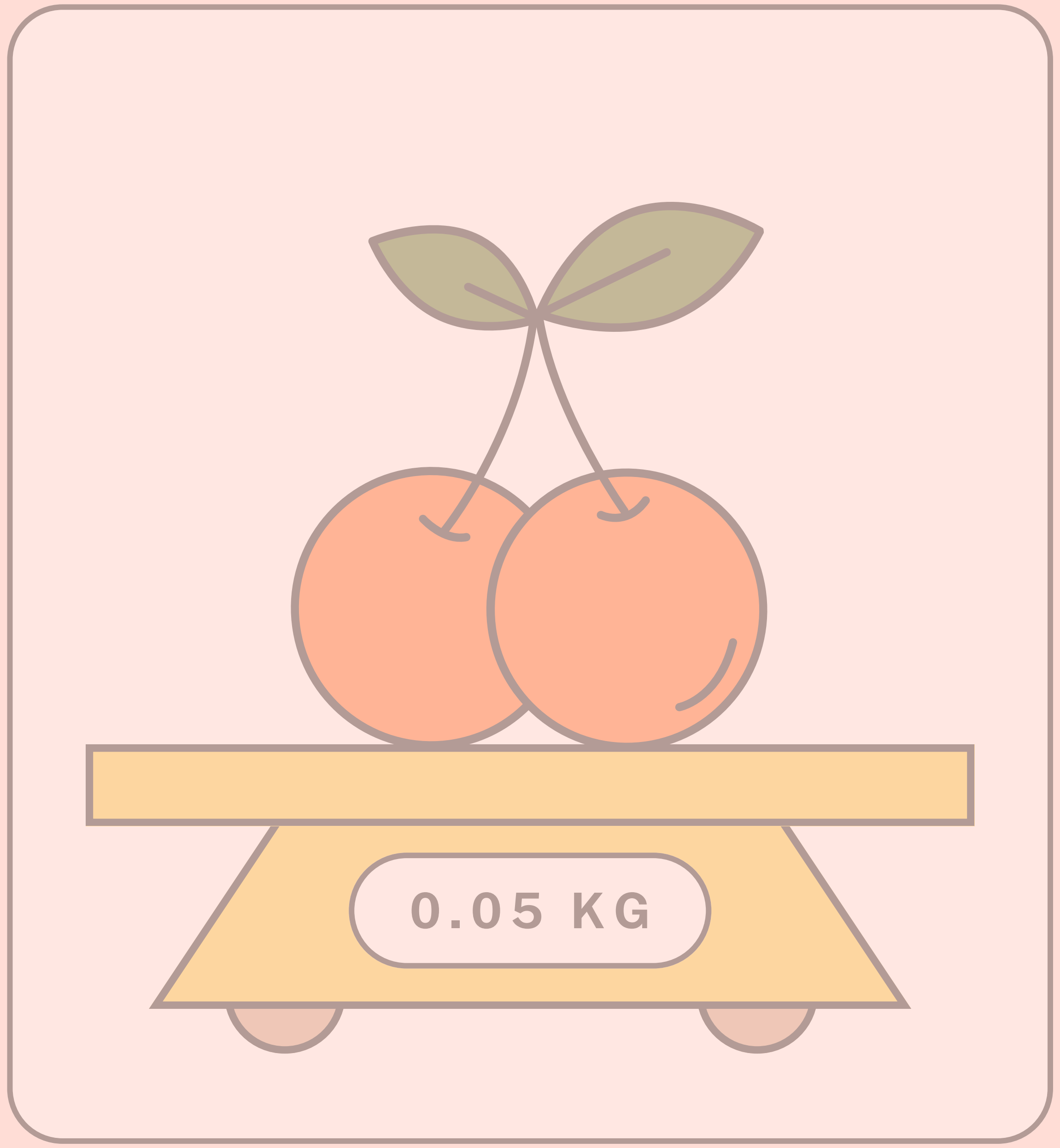
Which do you think weighs more?





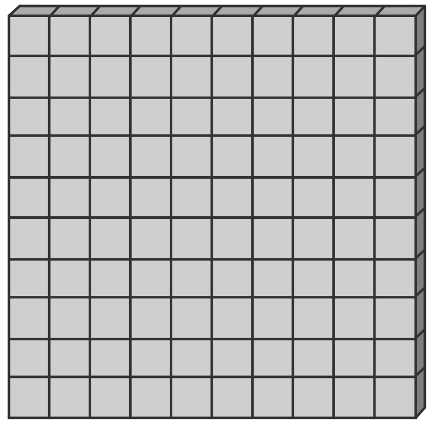
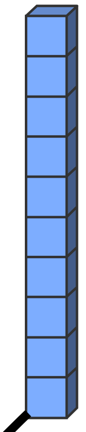
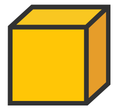
Which do you think weighs more?





The place value
of a digit is important
when comparing
decimal numbers.

REMEMBER THE VALUES IN DECIMALS:

Ones/Units	Decimal Point	Tenths	Hundredths
	.		

As the place value moves to the right, the value decreases by a tenth.

Divide by 10 to reduce <u>ones/units to tenths</u>.

Divide by 10 to reduce <u>tenths to hundredths</u>.

WHICH DECIMAL IS GREATER?

0.30 or 0.3

Ones/ Units	Decimal Point	Tenths	Hundredths
0	.	3	0

Ones/ Units	Decimal Point	Tenths	Hundredths
0	.	3	

They have the same value.

COMPARING USING PLACE VALUE CHARTS

STEP 1 Place the digits in their corresponding place value.

Another term to describe the ones place is units.

units	.	tenths	hundredths
5	.	8	7

units	.	tenths	hundredths
5	.	3	9

COMPARING USING PLACE VALUE CHARTS

STEP 2 Compare the digits from the largest place value.

A. If the digits are the same, move to the next place value to the right.

units	.	tenths	hundredths
5	.	8	7

units	.	tenths	hundredths
5	.	3	9

COMPARING USING PLACE VALUE CHARTS

STEP 2 Compare the digits from the largest place value.

B. If the digits are different, determine which digit is greater.

units	.	tenths	hundredths
5	.	8	7

units	.	tenths	hundredths
5	.	3	9

COMPARING USING PLACE VALUE CHARTS

STEP 3 Use the symbols, >, <, or = to describe the relationship of two numbers.

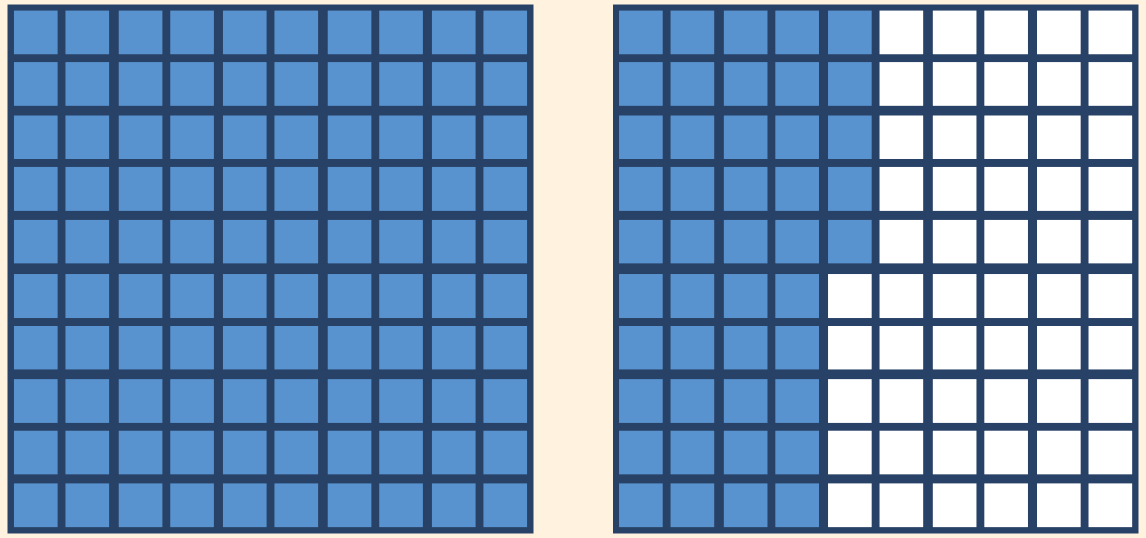
units	.	tenths	hundredths
5	.	8	7
5	.	3	9

In the tenths place, 8 is greater than 3. This means that 5.87 is greater than 5.39. In symbols, 5.87 > 5.39.

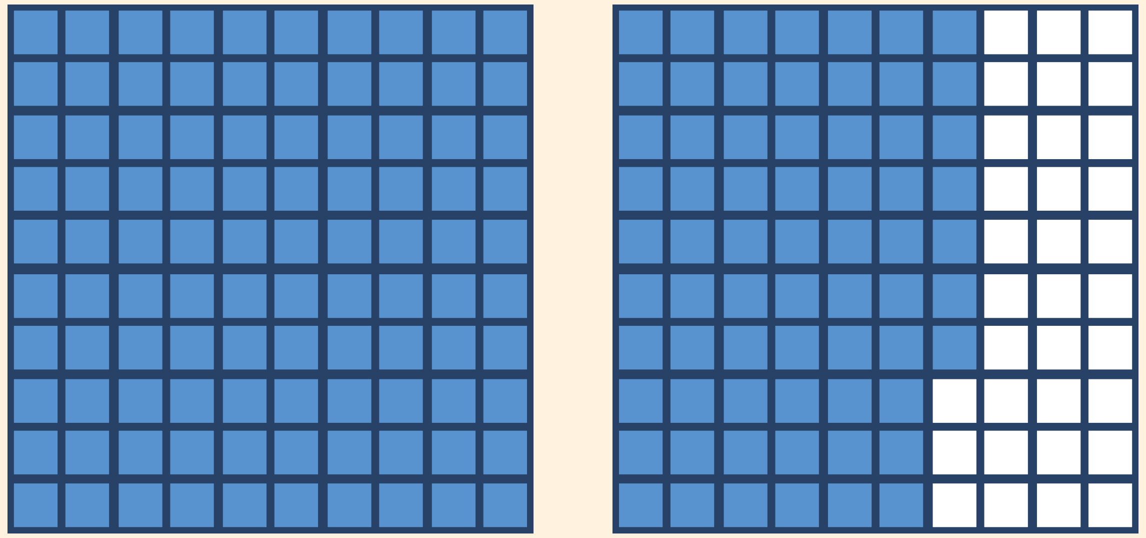
COMPARING USING AREA MODELS

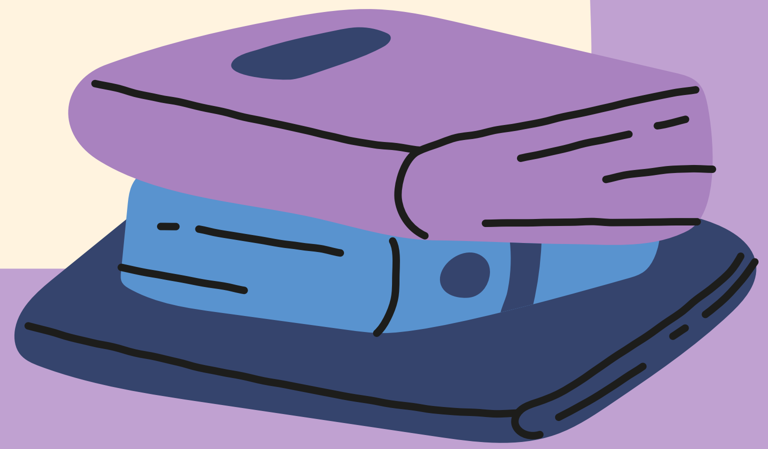
STEP 1 Draw a model for each decimal. Remember that 0.01 is 1 out of 100 parts and 1 is equivalent to 100 parts.

1.45



1.67

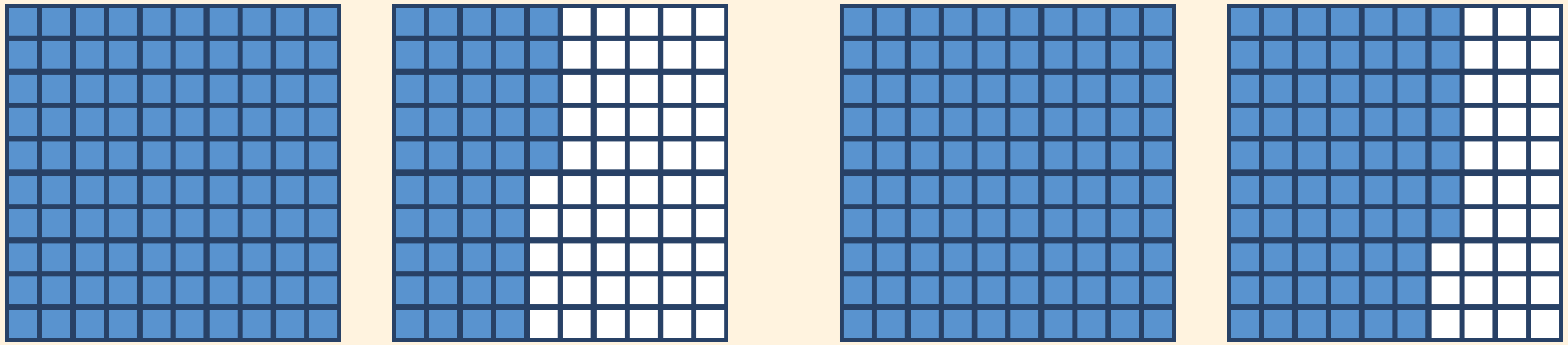




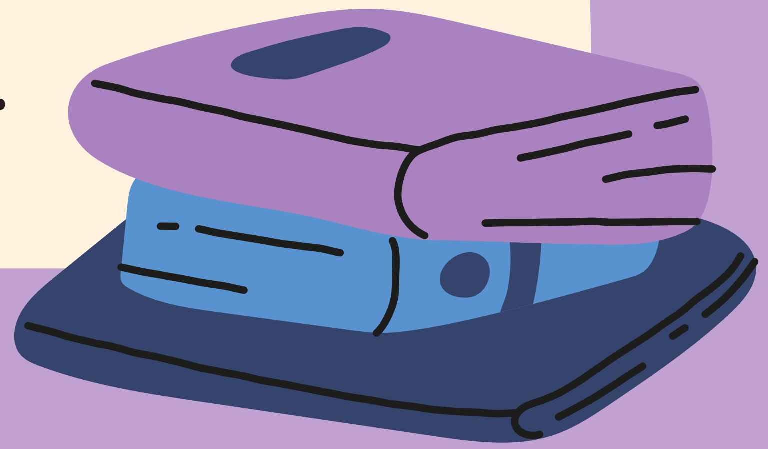
COMPARING USING AREA MODELS

STEP 2 Compare the shaded regions of each model. The number with more shaded regions has greater value.

1.45 | 1.67



The number with more shaded regions has greater value.

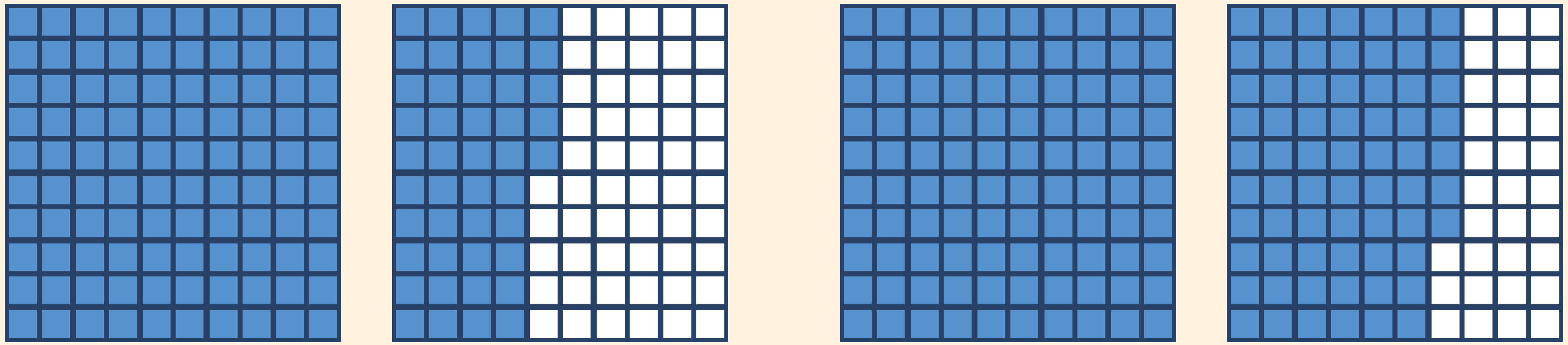


COMPARING USING AREA MODELS

STEP 3 Use the symbols, >, <, or = to describe the relationship of two numbers.

1.45

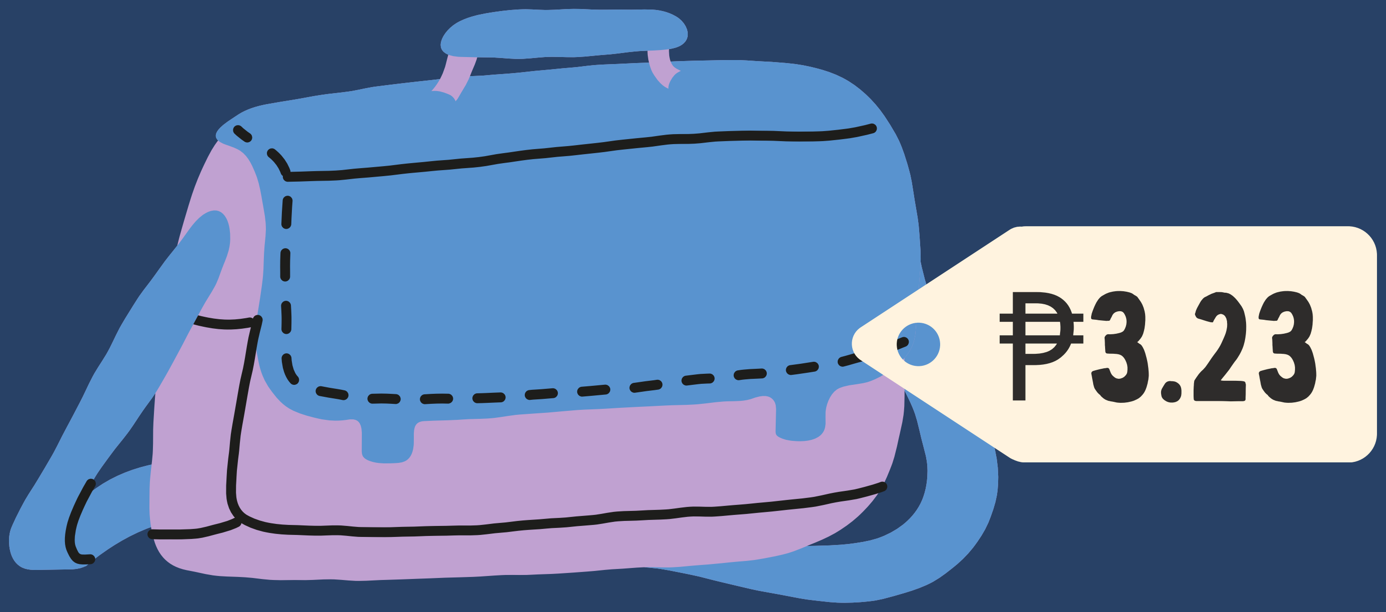
1.67



In symbols, 1.45 < 1.67.

WHICH BAG COMES WITH A LOWER PRICE TAG?

Place <, >, = on the blank to make the statement true.

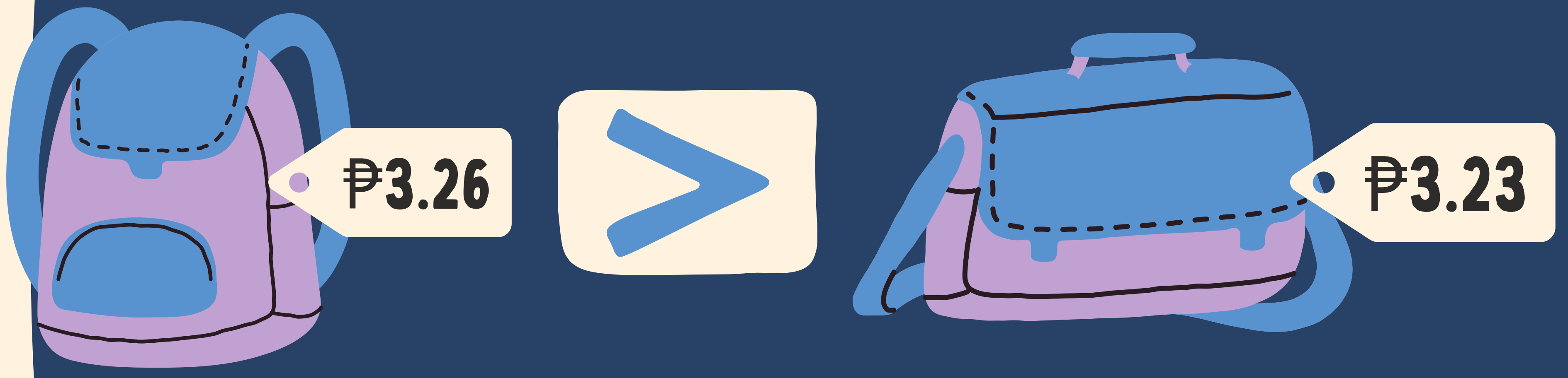


ANSWER KEY

WHICH BAG COMES WITH A LOWER PRICE TAG?

Place <, >, = on the blank to make the statement true.

units	.	tenths	hundredths
3	.	2	6
3	.	2	3



The satchel comes with a lower price tag.

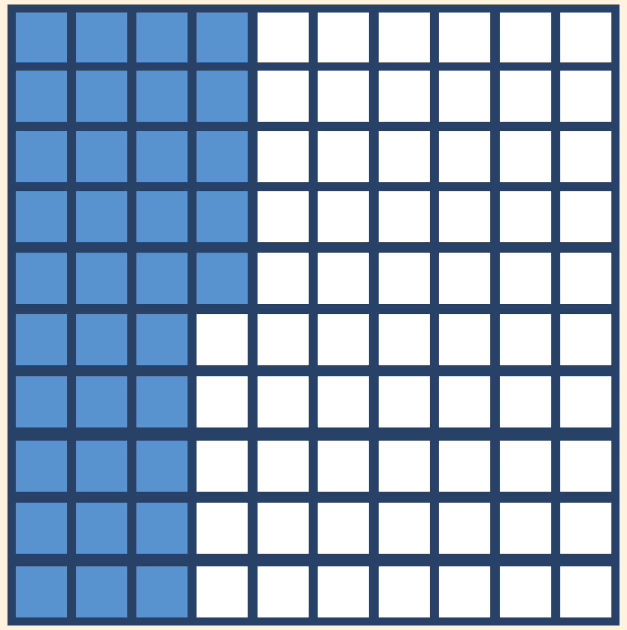
RECAP

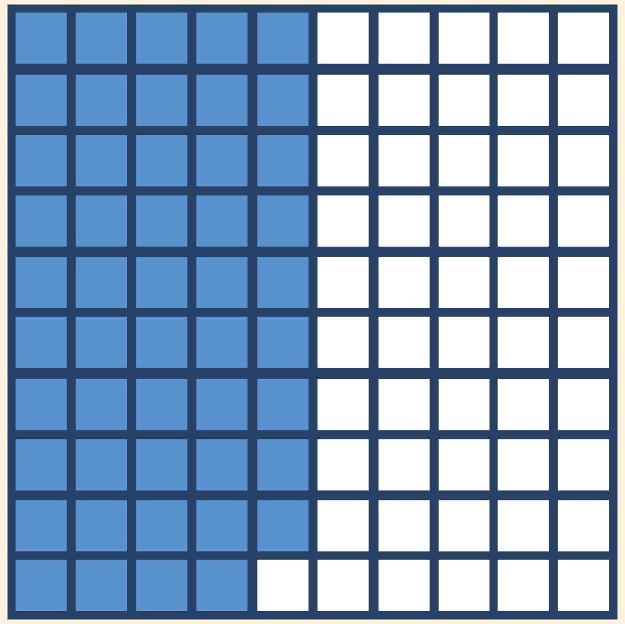
We can compare two decimal numbers using the following methods.

PLACE VALUE CHART

units	.	tenths	hundredths
3	.	2	6
3	.	2	3

AREA MODELS





If you only have £0.75, which snack can you buy?

If you only have £0.75, which snack can you buy?

Since £0.70 is less than £0.75.

WE CAN USE SYMBOLS TO SHOW THE RELATIONSHIP BETWEEN TWO NUMBERS.

$<$ means less than

$>$ means greater than

GREATER THAN OR LESS THAN?

0.8

?

0.76

<

or

>

GREATER THAN OR LESS THAN?

0.8 > 0.76

By comparing 8 and 7, we know that 8 is greater than 7.

ANSWER KEY

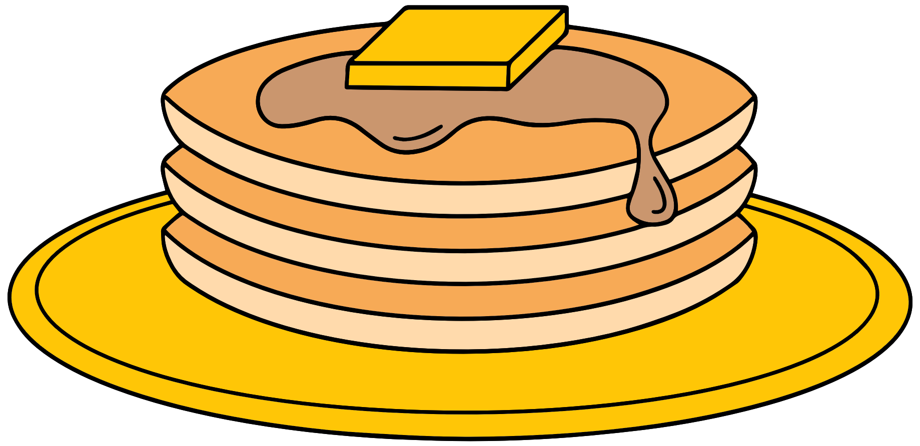
REMEMBER

Look at the place values, the further the digit is to the left, the greater is its value.

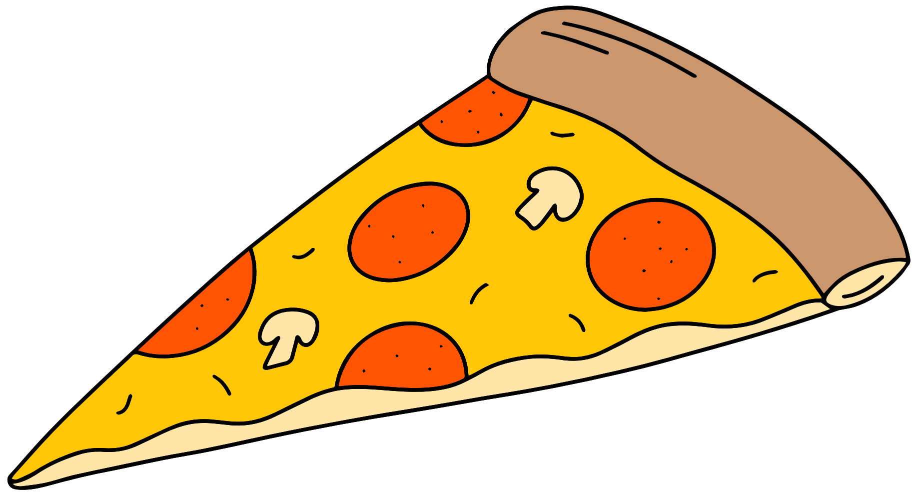
TRY THIS!

Which food would need the least amount of time to prepare?

1.25 HRS



1.2 HRS



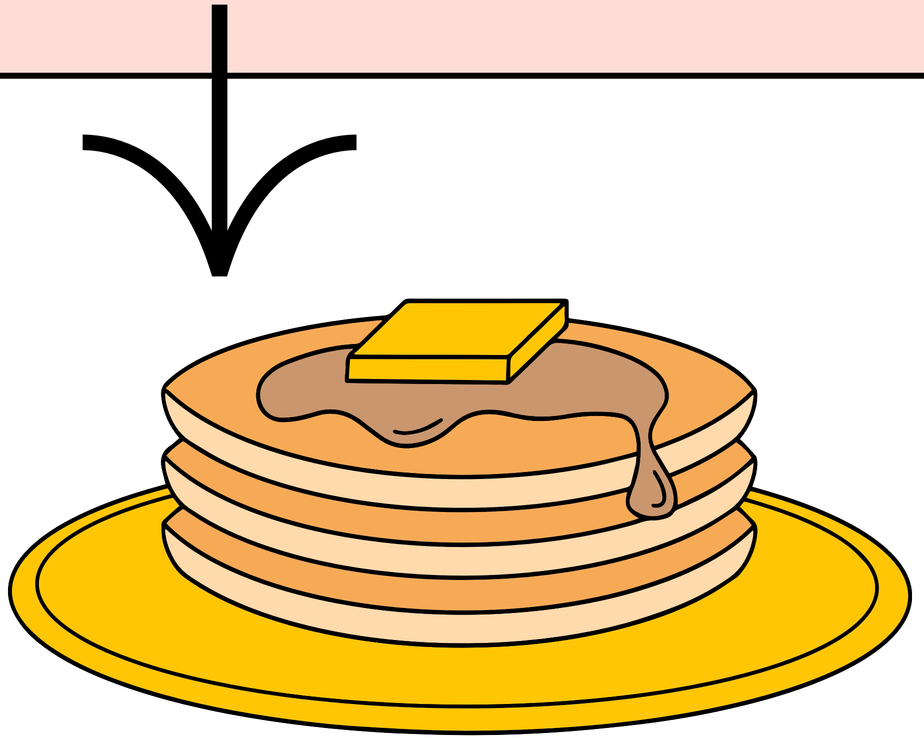
1.55 HRS

ANSWER

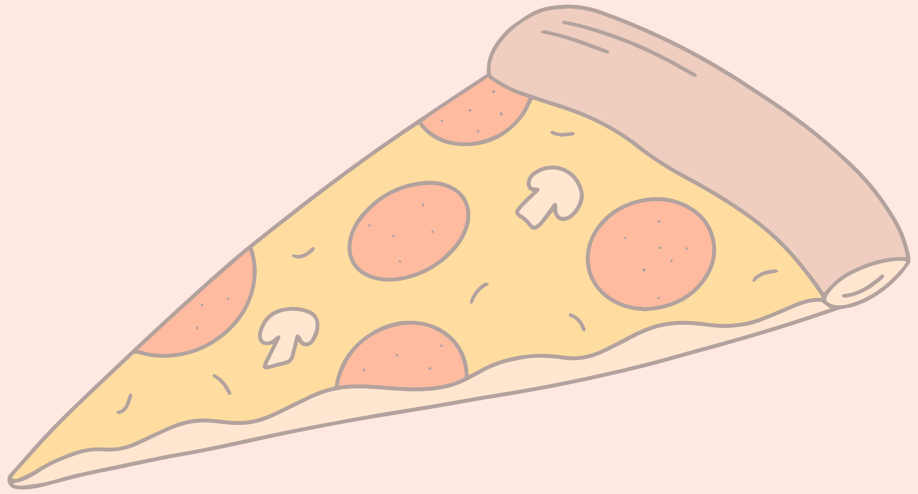
Which food would need the least amount of time to prepare?

Next is to arrange them in order, then share your answer in our next session!

1.25 HRS



1.2 HRS

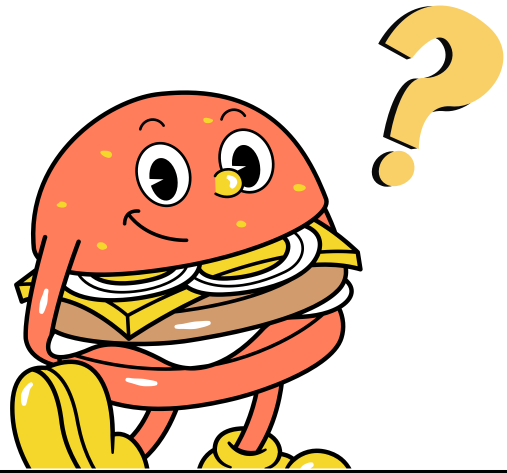


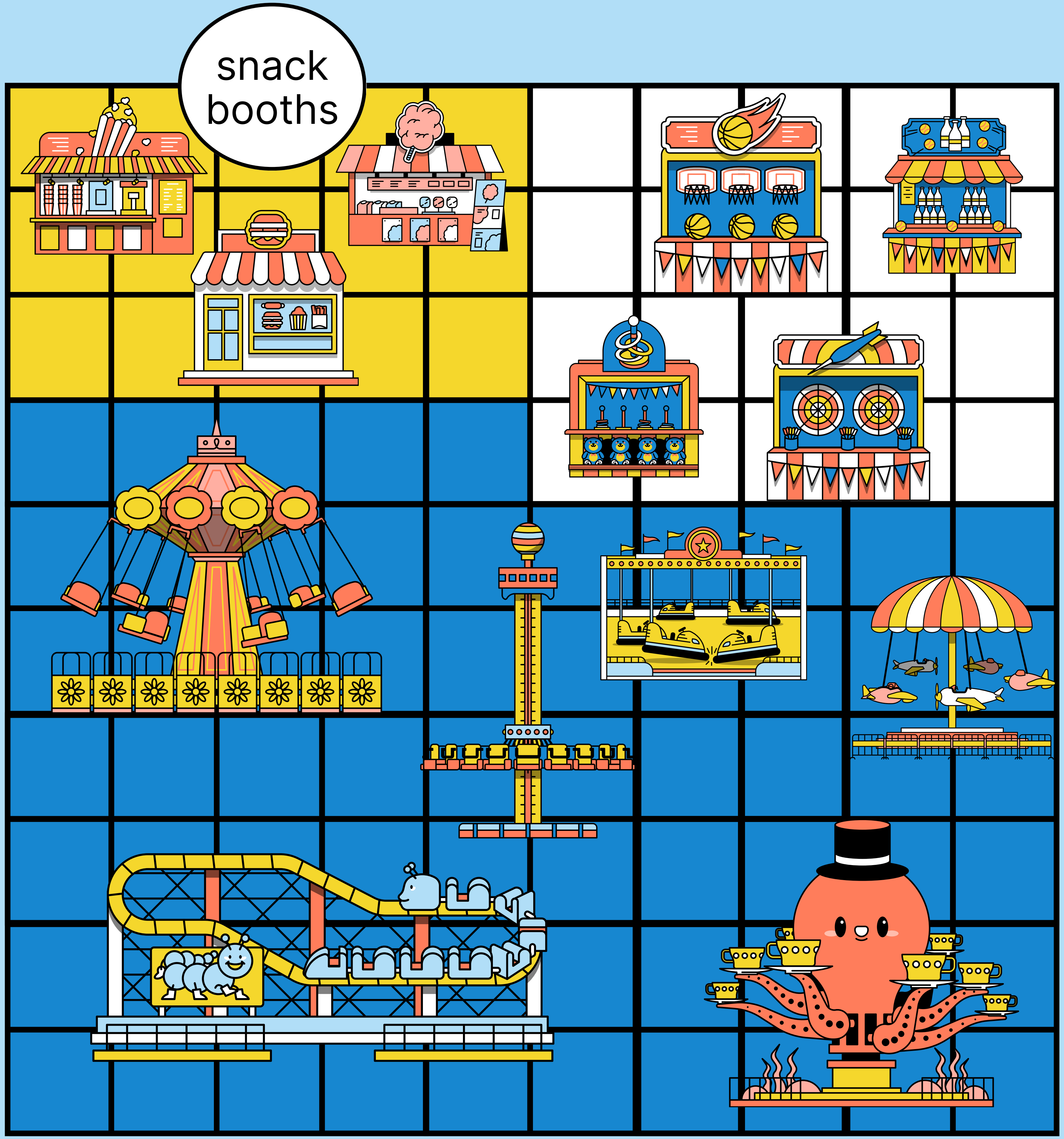
1.55 HRS

Investigate!

The snack booths take up **0.15** of the entire area of the amusement park.

What is that in fractions?

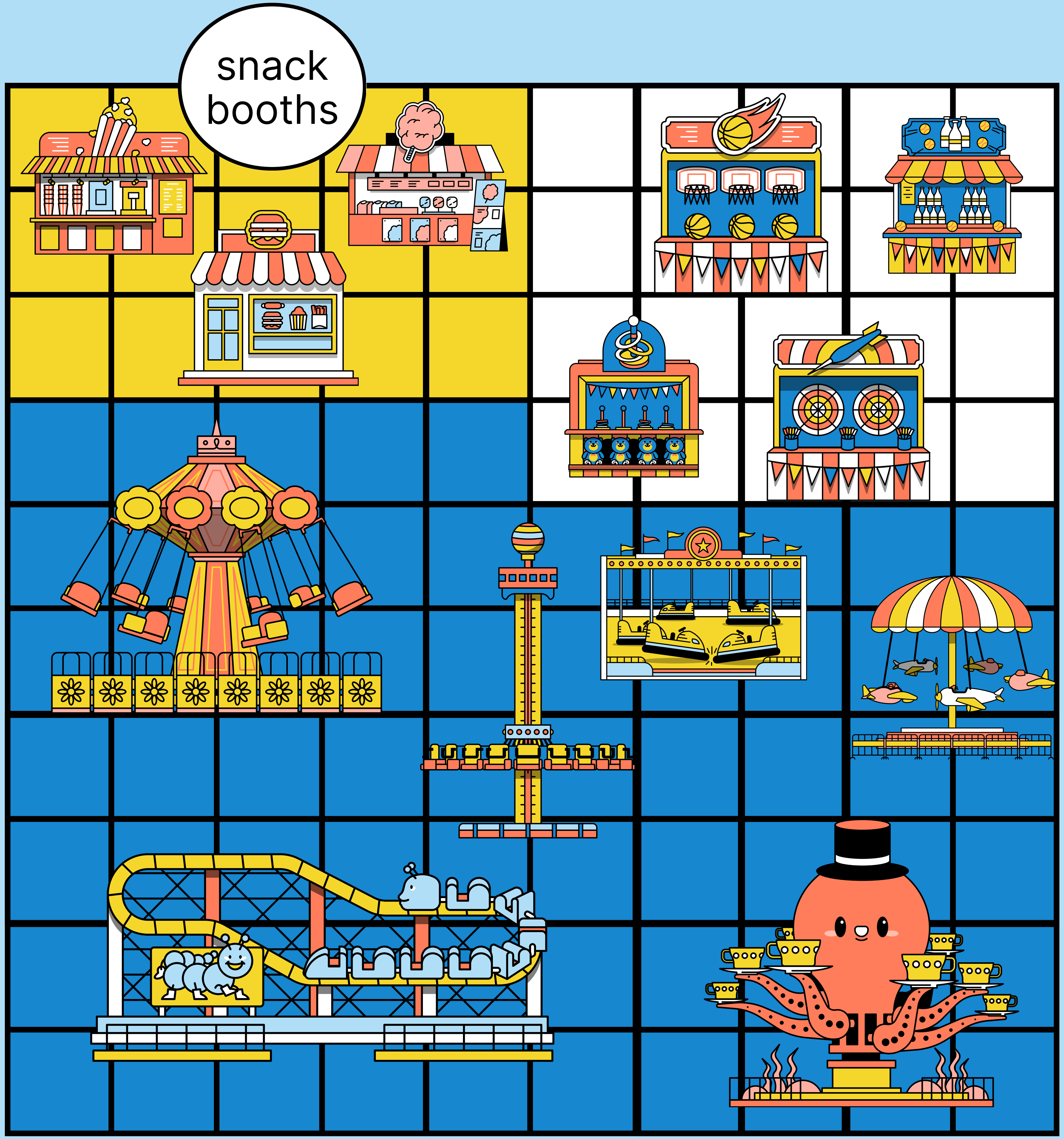




Answer:

We can answer the question by counting the squares that the snack booths take up, and that is

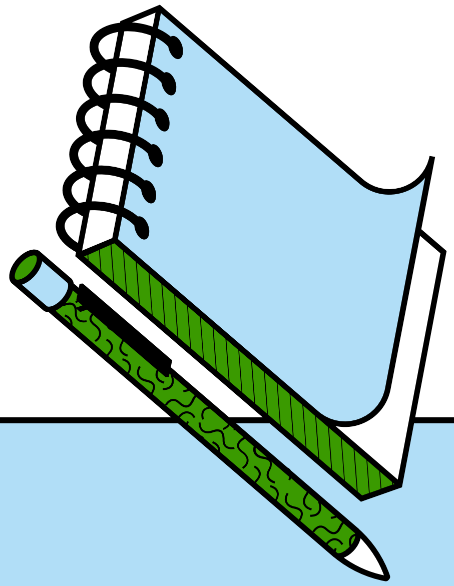
$$\frac{15}{100}$$



Converting Decimals to Fractions

Use the place value table to guide you on converting decimals to fractions.

Ones/Units	Decimal Point	Tenths	Hundredths	Thousandths
	.	$\frac{1}{10}$	$\frac{1}{100}$	$\frac{1}{1,000}$

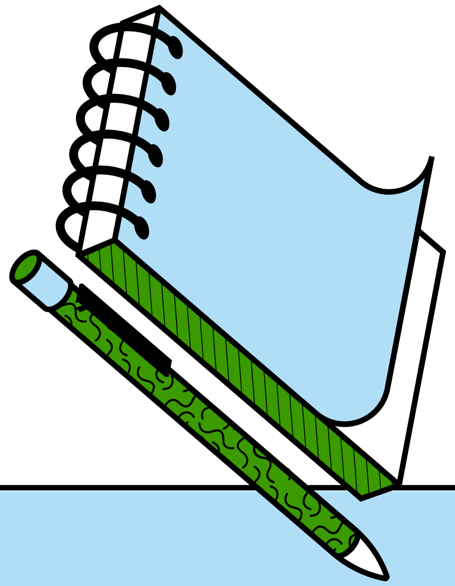


Note that 0.1 is $\frac{1}{10}$

Converting Decimals to Fractions

Now let's add our decimal 0.15 on the place value table.

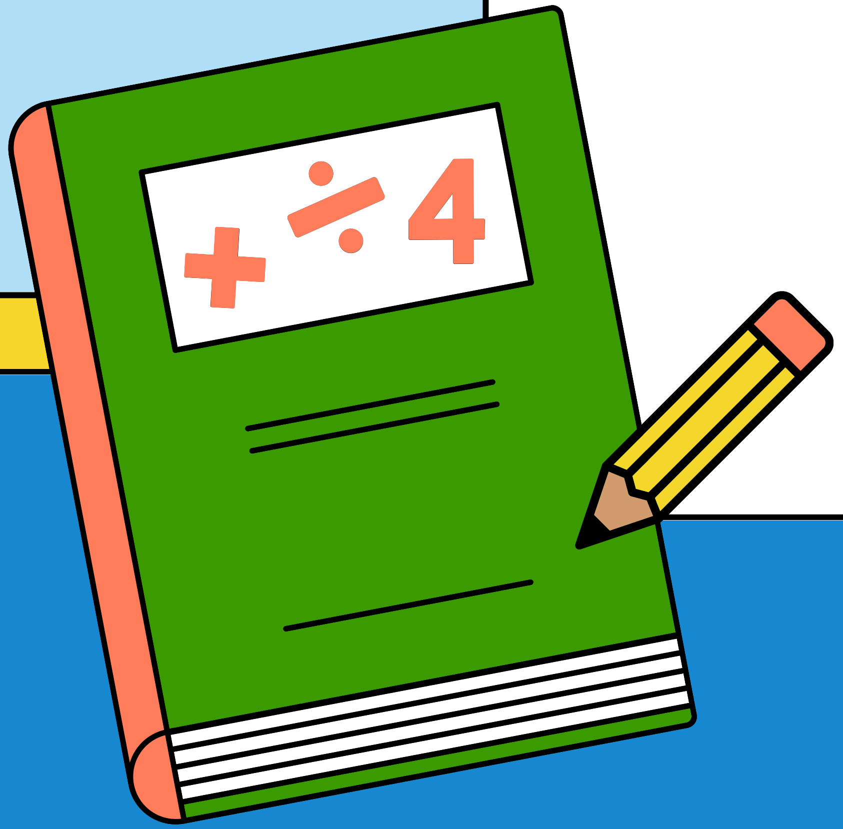
Ones/Units	Decimal Point	Tenths	Hundredths	Thousandths
0	.	1	5	



Since the decimal is up to the hundredths, its denominator will be 100, where 15 is the numerator.

Try this!

What is 3.4 in fractions? Simplify if possible.



Try this!

What is 3.4 in fractions? Simplify if possible.

Answer:

Since you have a whole number, the answer will be a mixed number, where 3 is the whole number and 4 tenths is in fraction.

$$3.4 = 3\frac{4}{10}$$

Converting Fractions to Decimals

It's easy to convert fractions that have 10, 100 or 1,000 as a denominator.

For $\frac{23}{100}$ since the denominator is 100, it means that the decimal will be up to the hundredths place.

Which means that, $\frac{23}{100} = 0.23$

Converting Fractions to Decimals

What if the denominator is not in the power of 10, like $\frac{3}{5}$?

STEP 1:

Think of a number that can be multiplied to the denominator to make it a power of 10.

With a denominator of 5, we can multiply it by **2**.

STEP 2:

Converting Fractions to Decimals

What if the denominator is not in the power of 10, like $\frac{3}{5}$?

STEP 3:

Write the fraction in decimals.

$$\frac{6}{10} = 0.6$$

THANK YOU!

Until next time...